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Studierichting: Informatica
Oefeningenreeks 1C nr. 11.22 [?]

$$z = \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

Berekening van de modulus ($r = |z|$):

$$r = \sqrt{a^2 + b^2}$$

$$r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$r = \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$r = \sqrt{1} = 1$$

Berekening van het argument (α):

$$\tan \alpha = \frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

$$\tan \alpha = \frac{-\sqrt{3}}{2} \cdot 2$$

$$\tan \alpha = \frac{-2\sqrt{3}}{2}$$

$$\alpha = \text{Bgtan}(-\sqrt{3})$$

$$\alpha = -\frac{\pi}{3}$$

De geometrische vorm ($z = r \cdot (\cos \alpha + i \sin \alpha)$):

$$z = \cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)$$

References

[1]